

Math Sec 1

1. Let n be the number obtained on rolling a fair die. If the probability that the system $x - ny + z = 6$
 $x + (n - 2)y + (n + 1)z = 8$
 $(n - 1)y + z = 1$ has a unique solution is $\frac{k}{6}$, then the sum of k and all possible values of n is:
(A) 21 (B) 24 (C) 20 (D) 22
2. Among the statements
(S_1) : If $A(5, -1)$ and $B(-2, 3)$ are two vertices of a triangle, whose orthocentre is $(0, 0)$, then its third vertex is $(-4, -7)$ and
(S_2) : If positive numbers $2a, b, c$ are three consecutive terms of an A.P., then the lines $ax + by + c = 0$ are concurrent at $(2, -2)$.
(A) only (S_2) is correct (B) only (S_1) is correct
(C) both are incorrect (D) both are correct
3. Let α, β be the roots of the quadratic equation $12x^2 - 20x + 3\lambda = 0, \lambda \in \mathbf{Z}$. If $\frac{1}{2} \leq |\beta - \alpha| \leq \frac{3}{2}$, then the sum of all possible values of λ is:
(A) 1 (B) 3 (C) 6 (D) 4
4. If $\lim_{x \rightarrow 0} \frac{e^{(a-1)x} + 2 \cos(bx) + (c-2)e^{-x}}{x \cos x - \log_e(1+x)} = 2$, then $a^2 + b^2 + c^2$ is equal to:
(A) 7 (B) 9 (C) 3 (D) 5
5. If $y = y(x)$ satisfies the differential equation $16(\sqrt{x}\sqrt{y} + 9x)(4 + \sqrt{9 + \sqrt{x}}) \cos y dy = (1 + 2 \sin y) dx, x > 0$ and $y(256) = \frac{\pi}{2}, y(49) = \alpha$, then $2 \sin \alpha$ is equal to:
(A) $2(\sqrt{2} - 1)$ (B) $\sqrt{2} - 1$
(C) $3(\sqrt{2} - 1)$ (D) $2\sqrt{2} - 1$
6. Let $P(10, 2\sqrt{15})$ be a point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$, whose foci are S and S' . If the length of its latus rectum is 8, then the square of the area of $\Delta PSS'$ is equal to:
(A) 900 (B) 4200 (C) 2700 (D) 1462
7. Let f and g be functions satisfying $f(x + y) = f(x)f(y), f(1) = 7$ and $g(x + y) = g(xy), g(1) = 1$, for all $x, y \in \mathbf{N}$. If $\sum_{x=1}^n \left(\frac{f(x)}{g(x)}\right) = 19607$, then n is equal to:
(A) 7 (B) 6 (C) 5 (D) 4
8. Let S and S' be the foci of the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$ and $P(\alpha, \beta)$ be a point on the ellipse in the first quadrant. If $(SP)^2 + (S'P)^2 - SP \cdot S'P = 37$, then $\alpha^2 + \beta^2$ is equal to:
(A) 13 (B) 15 (C) 17 (D) 11
9. Let $S = \{z \in \mathbf{C} : 4z^2 + \bar{z} = 0\}$. Then $\sum_{z \in S} |z|^2$ is equal to:
(A) $\frac{7}{64}$ (B) $\frac{1}{16}$
(C) $\frac{5}{64}$ (D) $\frac{3}{16}$
10. The number of elements in the relation $R = \{(x, y) : 4x^2 + y^2 < 52, x, y \in \mathbf{Z}\}$ is
(A) 67 (B) 77 (C) 86 (D) 89
11. Let C_r denote the coefficient of x^r in the binomial expansion of $(1 + x)^n, n \in \mathbf{N}, 0 \leq r \leq n$. If $P_n = C_0 - C_1 + \frac{2^2}{3} C_2 - \frac{2^3}{4} C_3 + \dots + \frac{(-2)^n}{n+1} C_n$, then the value of $\sum_{n=1}^{25} \frac{1}{P_{2n}}$ equals.
(A) 580 (B) 675 (C) 525 (D) 650
12. Let the locus of the mid-point of the chord through the origin O of the parabola $y^2 = 4x$ be the curve S . Let P be any point on S . Then the locus of the point, which internally divides OP in the ratio 3:1, is:
(A) $2x^2 = 3y$ (B) $2y^2 = 3x$
(C) $3y^2 = 2x$ (D) $3x^2 = 2y$
13. If $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ is a solution of the system of equations $AX = B$, where $\text{adj } A = \begin{bmatrix} 4 & 2 & 2 \\ -5 & 0 & 5 \\ 1 & -2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 4 \\ 0 \\ 2 \end{bmatrix}$, then $|x + y + z|$ is equal to:
(A) 2 (B) 1
(C) 3 (D) $\frac{3}{2}$
14. Let $[\cdot]$ denote the greatest integer function, and let $f(x) = \min\{\sqrt{2}x, x^2\}$. Let $S = \{x \in (-2, 2) : \text{the function } g(x) = |x| [x^2] \text{ is discontinuous at } x\}$. Then $\sum_{x \in S} f(x)$ equals
(A) $2 - \sqrt{2}$ (B) $1 - \sqrt{2}$
(C) $2\sqrt{6} - 3\sqrt{2}$ (D) $\sqrt{6} - 2\sqrt{2}$
15. Let $f(x) = [x]^2 - [x + 3] - 3, x \in \mathbf{R}$, where $[\cdot]$ is the greatest integer function. Then
(A) $\int_0^2 f(x) dx = -6$
(B) $f(x) = 0$ for finitely many values of x
(C) $f(x) > 0$ only for $x \in [4, \infty)$

(D) $f(x) < 0$ only for $x \in [-1, 3]$

16. Let L be the line $\frac{x+1}{2} = \frac{y+1}{3} = \frac{z+3}{6}$ and let S be the set of all points (a, b, c) on L , whose distance from the line $\frac{x+1}{2} = \frac{y+1}{3} = \frac{z-9}{0}$ along the line L is 7. Then $\sum_{(a,b,c) \in S} (a + b + c)$ is equal to:

(A) 40 (B) 28 (C) 34 (D) 6

17. The area of the region $A = \{(x, y) : 4x^2 + y^2 \leq 8 \text{ and } y^2 \leq 4x\}$ is:

(A) $\frac{\pi}{2} + 2$ (B) $\pi + 4$
(C) $\pi + \frac{2}{3}$ (D) $\frac{\pi}{2} + \frac{1}{3}$

18. If the mean deviation about the median of the number $k, 2k, 3k, \dots, 1000k$ is 500, then k^2 is equal to:

(A) 4 (B) 1 (C) 9 (D) 16

19. Let the domain of the function $f(x) = \log_3 \log_5 (7 - \log_2 (x^2 - 10x + 85)) + \sin^{-1} \left(\left| \frac{3x-7}{17-x} \right| \right)$ be $(\alpha, \beta]$. Then $\alpha + \beta$ is equal to:

(A) 8 (B) 9 (C) 12 (D) 10

20. Let $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$ and $\vec{b} = \lambda\hat{j} + 2\hat{k}, \lambda \in \mathbb{Z}$ be two vectors. Let $\vec{c} = \vec{a} \times \vec{b}$ and $\vec{d}$ be a vector of magnitude in yz -plane. If $|\vec{c}| = \sqrt{53}$, then the maximum possible value of $(\vec{c} \cdot \vec{d})^2$ is equal to:

(A) 208 (B) 26 (C) 52 (D) 104

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21. Let $\cos(\alpha + \beta) = -\frac{1}{10}$ and $\sin(\alpha - \beta) = \frac{3}{8}$, where $0 < \alpha < \frac{\pi}{3}$ and $0 < \beta < \frac{\pi}{4}$. If $\tan 2\alpha = \frac{3(1-r\sqrt{5})}{\sqrt{11}(s+\sqrt{5})}$, $r, s \in \mathbb{N}$, then $r + s$ is equal to ____.

22. Let S be the set of the first 11 natural numbers. Then the number of elements in $A = \{B \subseteq S : n(B) \geq 2 \text{ and the product of all elements of } B \text{ is even}\}$ is ____.

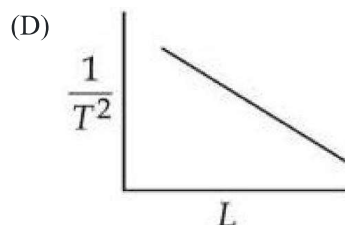
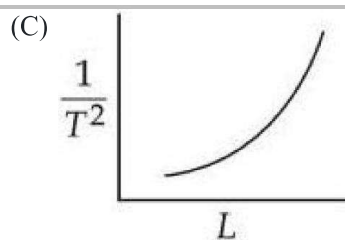
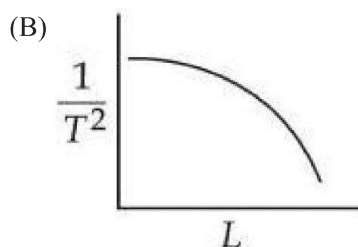
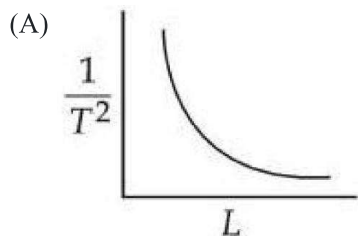
23. Let $[\cdot]$ be the greatest integer function. If $\alpha = \int_0^{64} (x^{1/3} - [x^{1/3}]) dx$, then $\frac{1}{\pi} \int_0^{\alpha\pi} \left(\frac{\sin^2 \theta}{\sin^6 \theta + \cos^6 \theta} \right) d\theta$ is equal to ____.

24. Suppose a, b, c are in A.P. and $a^2, 2b^2, c^2$ are in G.P. $a < b < c$ and $a + b + c = 1$, then $9(a^2 + b^2 + c^2)$ is equal to ____.

25. Let a vector $\vec{a} = \sqrt{2}\hat{i} - \hat{j} + \lambda\hat{k}, \lambda > 0$, make an obtuse angle with the vector $\vec{b} = -\lambda^2\hat{i} + 4\sqrt{2}\hat{j} + 4\sqrt{2}\hat{k}$ and an angle $\theta, \frac{\pi}{6} < \theta < \frac{\pi}{3}$ with the positive z -axis. If the set of all possible values of λ is $(\alpha, \beta) - \{\gamma\}$, then $\alpha + \beta + \gamma$ is equal to ____.

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26. Using a simple pendulum experiment g is determined by measuring its time period T . Which of the following plots represent the correct relation between the pendulum length L and time period T ?



27. Figure shows the circuit that contains three resistances 9Ω each and two inductors (4 mH each). The reading of the ammeter at the moment switch K is turned ON, is ____ A.